

Introduction To Database

ER Diagram

Case Study 1:

In a university management system, a university may have several faculties. However, a faculty belongs to only one university. A faculty can be identified by a faculty ID. The system also stores faculty name and phone number. A department can be part of only one faculty, but a faculty can have multiple departments. The unique property of each department is the department number. The system also stores the name and location of each department. A department may have multiple phone numbers. A professor belongs to only one department, but a department can have many professors. A professor can be identified by a professor ID. The system also stores professor name, hire date, and office location. A course can be offered by multiple departments, and a department can offer multiple courses. The unique property of each course is the course code. The system also stores the name and description of each course. A student can enroll in multiple courses, and a course can have multiple students. A student can be identified by a student ID. The system also stores student name, enrollment year, and email address.

Case Study 2:

In a healthcare management system, a hospital may have multiple departments. Each department can be identified by a department ID. The system also stores the department name and location. A doctor can work in multiple departments, and a department can have multiple doctors. A doctor can be identified by a doctor ID. The system also stores the doctor's name, specialization, and contact number. A patient can be treated by multiple doctors, and a doctor can treat multiple patients. A patient can be identified by a patient ID. The system also stores the patient's name, date of birth, and contact number. A treatment session is associated with one doctor and one patient. The unique property of each treatment session is the session ID. The system also stores the date, time, and notes of the treatment session. A hospital may also have multiple medical tests. Each test can be identified by a test ID. The system also stores the test name, description, and cost. A patient can undergo multiple medical tests, and each test can be taken by multiple patients. The system should track which tests each patient has undergone, including the date of the test and the result.

Case Study 3:

An author is going to write a book. The author has a name, address, and phone number. Address has three parts: city, road and house. Also, the author has a specific ID number. Book has title, year, price, and unique ISBN number. At a time, one author can write one book. When an author writes a book, then it will need a publisher. Because a book needs to be published through a publisher. Publisher has name, address, and email address. Email is a publication mail. That's why this mail can't be used by others those are not from publication. The address has three parts: city, zip and house no. Also, a book will be published in a specific publication. But a publication will publish one or more books. After publishing the book, it will be stocked in a warehouse. The warehouse has a specific code, address, and phone number. One book has multiple copies. So, the same type of copies can be stocked in a different warehouse.

Case Study 4:

A car insurance company whose customers own one or more cars each. Each car has been associated with it zero to any number of recorded accidents. Each insurance policy covers one or more cars and has one or more premium payments associated with it. Each payment is for a particular period of time and has an associated due date, and the date when the payment was received.

Case Study 5:

You are asked to create a database to produce a report on customer details. The report is to list the customer's name, account balance, credit limit, and other customer details. The report will also list the customer account rep (one of our sales employees). Our sales reps manage many customers each, but each customer will be managed by only one account rep at any one time. However, your design should allow customers to be managed by many account reps as it is possible that some of our employees may leave the company - thus, requiring new account reps for a customer.

Case Study 6:

A company operates a warehouse parts supply business. The company has several warehouses located in Toronto which each store several hundreds of automotive parts. We need to keep a record of how many parts are "on hand" - meaning inventory levels that tell us how many we have for each part. To help us organize our parts, each part is assigned a specific classification. There are 4 classifications that we use to organize hundreds of parts.

Case Study 7:

A large bank operates in several divisions. Information Technology (IT) is operated as one of these divisions. Within the IT division there are many departments that are managed by one manager, and all IT employees belong to one of these departments. The IT division assigns its employees to one or more ongoing projects in the bank. A project may be planned, but not have any employees assigned to it for several months. Each project will have a single employee assigned who acts as a project leader.

Query Writing

- 1) Display the ename, job, hiredate, and empno for each employee, with empno appearing first. Provide an alias "STARTDATE" for the HIREDATE column.
- 2) Display all unique jobs from the EMPLOYEES table.
- 3) The HR department wants more descriptive column headings for its report on employees. Name the column headings Emp # for empno, Employee for ename, Job for JOB, and Hire Date for HIREDATE, respectively. Then run your query.
- 4) Display the ename concatenated with the salary (separated by a comma and space) and name the column Employee and Title.
- 5) Create a query to display all the data from that table. Separate each column output by a comma. Name the column title THE_OUTPUT.
- 6) List the deptno from emp uniquely.
- 7) Show the information of all employees like "SMITH go salary \$800".
- 8) Check whether all the empno are indeed unique.
- 9) List "Annual salary" for all employee use column aliasing.
- 10) List the information for all employees whose job is either manager or analyst.
- 11) List the department name whose department no is 20.
- 12) Count the number of employees whose salary is more than \$800 and whose name contains more than 10 characters. Display the result with an appropriate label.
- 13) Create a query to display the name, salary, commission of employees earning more than \$2850.
- 14) Create a query to display the employee's name, salary and department number for employee number 7566.
- 15) Display the name and salary for all employees whose salary is not in the range of \$1500 and \$2850
- 16) Display the names of all employees where the third letter of their name is A.
- 17) Display the name of all employees who have two L in their name and are in department 30 or their manager is 7782.
- 18) Display the name, job, and salary for all employees whose job is clerk or analyst, and their salary is not equal to \$1000, \$3000, or \$5000.
- 19) Display the name, salary, and commission for all employees whose commission amount is greater than their salary increased by 10%.
- 20) Create a query to display the employee's name, job title, and length of service in months for employees who were hired before January 1, 1985, and who do not have a manager. Order the results by length of service in descending order.
- 21) Display the employees' name, job, hire date, and department number for employees who were hired in the first quarter (January to March) of any year and are in departments 10 or 20. Order the results by hire date in ascending order.
- 22) Write a query to list the name, job title, salary, and commission of employees who earn a commission greater than 500. Order the results by salary in descending order and commission in descending order.

- 23) Create a query to display the employees' name, department number, and the length of time in years they have been with the company (using the current date) for employees who have been with the company for more than 5 years.
- 24) Display the employee's name, job, and hire date of employees who were hired between January 1, 1980, and December 31, 1982, and are in department 10 or 40. Order the results by hire date in descending order and then by name in ascending order.
- 25) Create a query to display the employee's name, job title, and annual salary (assuming 12 months in a year) for employees who earn more than \$2500 per month and are in department 20 or 30. Label the columns as 'Employee', 'Job', and 'Annual Salary'.
- 26) Write a query that will display name, hire date who are hired from '17-dec-80'to '20-jul-81'.
- 27) Display the name, hire date, updated salary including a 10% bonus (calculated as 10% salary plus commission if available, otherwise just salary), and the day of the week for employees who were hired on a Sunday in February. Ensure the updated salary is formatted with the local currency symbol and proper grouping for thousands. Sample output-

ENAME	HIREDATE	Updated Salary	Date
WARD	22-FEB-81	\$0001,925.000	SUNDAY

- 28) Write a query to display the current date and calculate the date of the first Tuesday that occurs four months after the current month (round the month). Label the column as ROUNDED_MONTH. Sample output-

SYSDATE	ROUNDED_MONTH
22-NOV-24	08-APR-25

- 29) Display the employee's name and their total earnings, which is the sum of their salary and commission (if any). Label the column as Total Earnings:
- 30) Display the employee's name when the name has the character 'R' in it and the number of years they have been working, rounded to two decimal places. Assume today's date is July 1, 2024.
- 31) Display the average salary for each job title, rounded to the nearest whole number. Label the column as Average Salary.
- 32) Display the employee's name, job title, and hire date for employees hired after employee number 7900, ordered by hire date:
- 33) Display the employees' names and the number of days since they were last reviewed. Assume today's date is July 1, 2024, and their last review was exactly one year ago:
- 34) Write a query to count the number of unique hire dates. Replace null values with "No Hire Date Provided". If the counted amount is equal to 5, return "Null value counted"; otherwise, return "Null value not counted". Rename the resultant column as "HIRE INFORMATION".
- 35) Display the employee's name and hire date in the format "YYYY-MM-DD".
- 36) Display the hire date in a readable format and calculate years worked. Sample output-

Employee Name	Formatted Hire Date	Years Worked
SMITH	Wednesday, 17-December -1980	43.93

- 37) Show the employee's name and formatted salary with a dollar sign and commas separating thousands.
- 38) Write a query to display the employee's name and the commission as a number. If commission is null, show 0 instead.
- 39) Write a query to display the employee's name, job title, and hire date in the format: Day, DD Month YYYY
- 40) Display the employee's name, hire date, and the exact day of the week they were hired.
- 41) Create a query to display the employee's name, current salary, and their salary in words (e.g., "One Thousand Five Hundred Dollars").
- 42) Find the name of the employee whose hire date is today's date.
- 43) Display salary, calculate 10% bonus, and display in this format. Sample Output-

Employee Name	Current Salary	Salary With Bonus
SMITH	\$800.00	\$880.00
ALLEN	\$1,600.00	\$1,760.00

- 44) Write a query to show the name, department, and the number of years they have worked, rounded to one decimal place.
- 45) Convert a numeric value back into a date. For sample value 20240701, sample output-

Converted Date
01-JUL-24

- 46) Write a query to show the name of employees whose hire date falls on a Sunday.
- 47) Find out job group having highest amount of total salary (Sal + comm).
- 48) Find the total number of employees and the average salary in each department.
- 49) Find the job category with the average salary.
- 50) Find the department with the highest total salary (including commission).
- 51) Find the total employee of every department.
- 52) Find the department with the lowest average salary.
- 53) Find employees who do not have a manager.
- 54) Find the employees who earn a commission higher than their salary.